

Lab Title

Binary Search Tree (BST) Implementation and Operations

Lab Objectives

Students will learn:

- What is a Binary Search Tree
- How BST differs from a normal Binary Tree
- How to insert nodes in BST
- How to search values in BST
- Tree traversals: Inorder, Preorder, Postorder
- Find minimum and maximum value
- Delete a node from BST
- Solve practice problems using BST

Part 1: Basic Concept

A **Binary Search Tree** is a binary tree where:

- Left child contains values **less than** parent node
- Right child contains values **greater than** parent node
- Each node can have maximum two children

Example:

```
    50
   /  \
  30   70
 /  \  /  \
20 40 60 80
```

In this tree:

- 30 is less than 50, so it is on left side
- 70 is greater than 50, so it is on right side

Part 2: Basic BST Implementation

```
1. #include <iostream> using namespace std;

struct Node {
    int data;
    Node* left;
    Node* right;

    Node(int value) {
        data = value;
        left = right = NULL;
    }
};

Node* insert(Node* root, int value) {
    if (root == NULL) {
```

```

        return new Node(value);
    }

    if (value < root->data) {
        root->left = insert(root->left, value);
    } else if (value > root->data) {
        root->right = insert(root->right, value);
    }

    return root;
}

void inorder(Node* root) {
    if (root == NULL)
        return;

    inorder(root->left);
    cout << root->data << " ";
    inorder(root->right);
}

int main() {
    Node* root = NULL;

    root = insert(root, 50);
    insert(root, 30);
    insert(root, 70);
    insert(root, 20);
    insert(root, 40);
    insert(root, 60);
    insert(root, 80);

    cout << "Inorder Traversal: ";
    inorder(root);

    return 0;
}
2.

```

Expected output:

Inorder Traversal: 20 30 40 50 60 70 80

Lab Tasks

Task 1

Create a BST and insert the following values:

45, 25, 65, 15, 35, 55, 75

Print the tree using:

- Inorder traversal
- Preorder traversal
- Postorder traversal

Task 2

Write a function to search a value in BST.

Example:

Search 35 → Found

Search 90 → Not Found

Task 3

Write a function to find the minimum value in BST.

Hint:

Minimum value is found at the leftmost node.

Task 4

Write a function to find the maximum value in BST.

Hint:

Maximum value is found at the rightmost node.

Task 5

Count total number of nodes in a BST.

Example:

Total Nodes = 7

Task 6

Count leaf nodes in BST.

A leaf node has no left and no right child.

Task 7

Find height of BST.

Example:

Height = 3

Task 8

Delete a node from BST.

Handle all three cases:

1. Node has no child
2. Node has one child
3. Node has two children

Home Assignment

Question 1

Create a BST from the following values:

100, 50, 150, 30, 70, 120, 170, 20, 40, 60, 80

Perform:

- Insert
- Search
- Inorder traversal
- Minimum value
- Maximum value

Question 2

Write a program to print only even numbers from BST using inorder traversal.

Question 3

Write a program to calculate the sum of all nodes in BST.

Question 4

Write a program to count nodes greater than a given value k.

Example:

k = 50

Output: 5

Question 5

Write a program to check whether a given tree is a valid BST or not.